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Evaluation of Serum Electrolyte Abnormalities in Children with Malnutrition and Acute Diarrheal Disease

A Dissertation Submitted Towards Academic Fulfillment for the Award of

Degree of

DOCTOR OF PHARMACY

TO

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTAPUR

ANANTHAPURAMU

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PREFACE

Malnutrition is a major problem Globally It interacts with diarrhoea in a vicious circle lead to major complications in children. understanding the interplay between malnutrition and diarrhoea is crucial in pediatric healthcare, particularly in resource -limited settings where these conditions often co-exist Malnourished children are not only more susceptible to diarrheal illness but also face greater risks of serum electrolytes disturbances, which can exacerbate their clinical course and lead to serious complications.

This assessment seeks to explore the intricate relationship between malnutrition, diarrhoea and serum electrolytes disturbances in pediatric patients.

Our study aim to assess the frequency of serum electrolytes disturbances in malnourished children with diarrhoea.

We wish to express our main objective of our research work to assess the serum electrolytes disturbance in malnourished children with diarrhoea in pediatrics. The present study attempts to provide the information about the anthropometric measurements of the malnourished children and to provide information regarding nutritional supplement of the malnourished children.

We expect that our thesis information may be helpful for physicians, clinical pharmacists, other health care professionals and patient Guardians. we hope that this work will fulfill your expectations.

1.1 INTRODUCTION

The term, gastrointestinal tract (GIT), refers to the alimentary tract extending from the mouth to the anus. It is divided into mouth, oropharynx, esophagus, stomach, small intestine (jejunum and ileum) and large intestine (colon). On ingestion of food by the mouth, it is moved by the oropharynx into the esophagus. The latter acts as a conduit for transfer of food to the stomach where it is stored and mixed prior to its controlled passage into the small intestine where it is digested and absorbed. Then, it moves to large intestine where salts and water are conserved prior to excretion as feces. Undoubtedly, the normal GIT function is the net result of combined action of many functional systems. If there is a breakdown of any one, intestinal function will be disturbed. For instance, diarrhea develops when there is an enhanced overload of fluids from small intestines into the colon following maldigestion or active secretion, or when the absorptive capacity of the colon is compromised by disease.

A defect of intestinal mucosal immunity may lead to recurrent enteric infections. Intestinal obstruction follows loss of normal intestinal motility. An insult to the digestive or absorptive capacity of the GIT may cause digestive and abdominal complaints, failure to thrive and even weight loss [1].

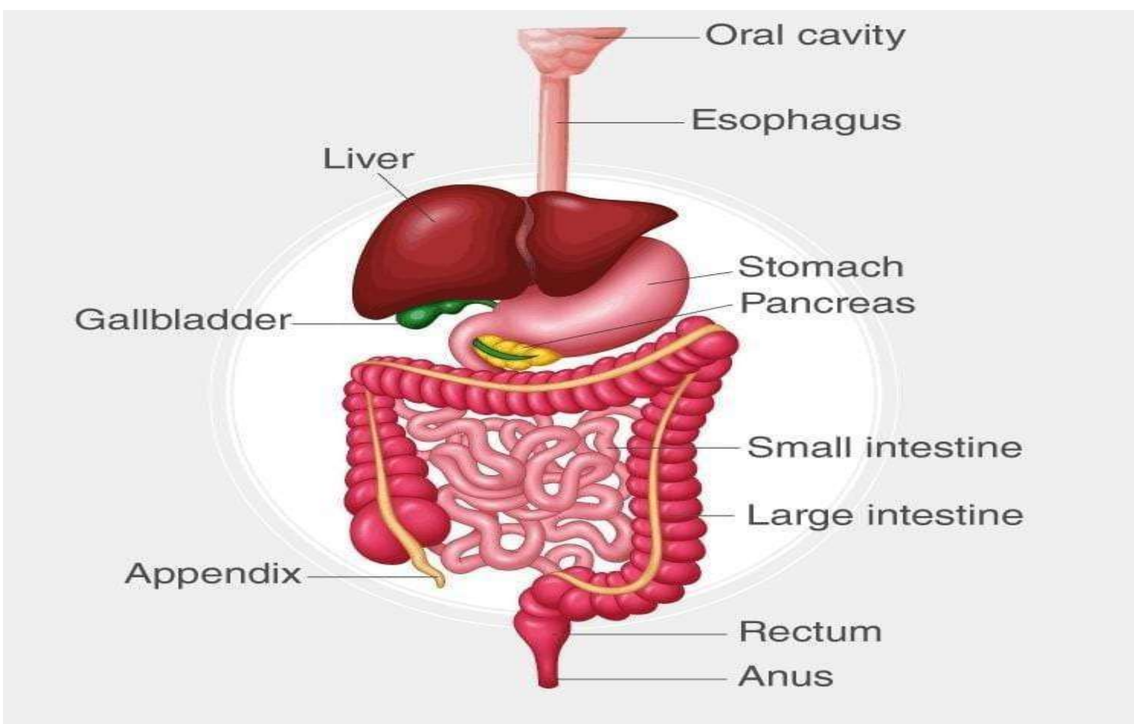

Fig NO. 1.1: GASTRO INTESTINAL SYSTEM

1.1.1 **SPECIAL INVESTIGATIVE WORK-UP FOR GASTROINTESTINAL DISORDERS**

For esophageal structure and function:

• Barium meal studies for defining anatomy of upper GIT and detecting advanced mucosal lesions, e.g. Varices and GER.

• Endoscopy for varices.

• 24-hour pH monito[redacted]s the most sensitive test for gastroesophageal reflux (GER).

• Esophagoscopy for esophagitis and mucosal biopsy.

For maldigestion/malabsorption

• Stool examination, including fat globules, reducing substances, pH, and microscopy. In case of a strong suspicion of intestinal parasitosis, it is advisable to carry stool microscopy by concentration method for at least 3 (preferably 6) consecutive days since ova and cysts frequently pass intermittently.

• 24-hour stool fat by fat balance studies and chemical examination of stools or by a semi-quantitative method termed "steatocrit". A daily stool fat of >5 g is considered indicative of steatorrhea.

•D-xylose test consists in measuring excretion of xylose in a 5-hour sample of urine after administering the pentose in a dose of 1 g/kg BW. An excretion of < 20 % points to malabsorption. A tolerance test too is available for infants and small children in whom collection of urine sample is quite cumbersome.

• Lactose tolerance test.

• Breath test involving measurement of H+.

• Barium meal follow–through, employing a [redacted]flocculable medium, may reveal intestinal changes indicative of malabsorption such as intestinal dilatation, flocculation, and atypical mucosal pattern, plus anatomic defects.

• Peroral or[redacted]doscopic gastric or jejunal biopsy. The jejunal biopsy provides vital histologic details as well as the material for enzymes, disaccharidases, especially, lactase, assay. It may also identify such pathogens as L. giardi[redacted] H. pylori. Gastric biopsy may be employed for histopathology, culture or rapid urea test for H. pylori.

• Schilling test measures the vitamin B12 absorption from the gut. It consists in administering a tracer dose of radioactive vitamin B12, after saturating body stores with vitamin B12, and its urinary excretion measured over the next 24 hours [1]. An excretion of < 5% indicates defective absorption from the ileum.

• Sweat chloride estimation by iontophoresis, using pilocarpine, is important for assay of the exocrine pancreatic function. A level of > 60 mEq/L usually established the diagnosis of cystic fibrosis.

1.2 <u>DIARRHEA: -</u>

Diarrheal diseases rank among the "top three" causes of death in pediatric population of the developing world. Globally, appr[redacted]tely 4-5 million deaths occur as a result of diarrheal diseases every year. Eight out of these 10 deaths are in the first 2 years of life, the most susceptible period for malnutrition. diarrheas account for about 20% of the hospitalized pediatric cases in India.

On an average, a child suffers from around 12 episodes of diarrhea, 4 such episodes occurring during the very infancy (first year). Existence of malnutrition makes the child very much vulnerable to d[redacted]al disease. It is estimated that incidence of diarrhea in malnourished children is 5 to 7 times higher than in healthy children. Likewise, its severity too is 3 to 4 times greater [1].

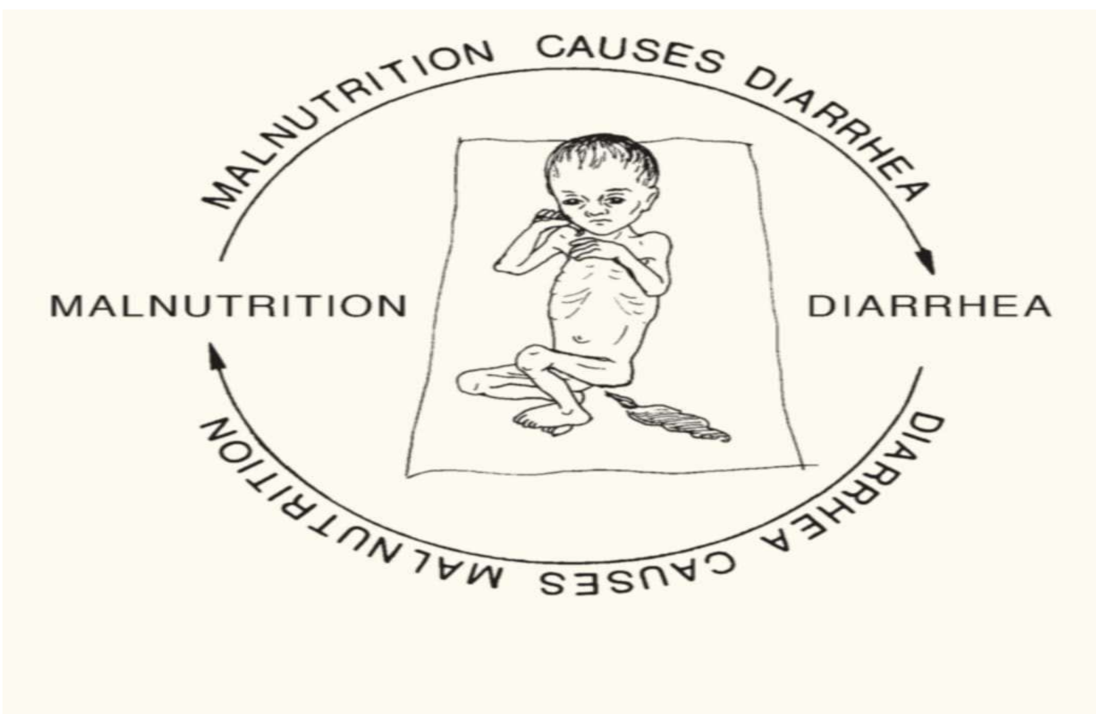

Fig NO. 1.2: Malnutrition with Diarrhea

Diarrhea means passage of 3 or more loose or watery motions per 24 hours, resulting in excessive loss of fluid and electrolytes in stools. Secretory, osmotic or motility abnormalities, singularly or in combination, form the basis of all diarrheal episodes. Secretory diarrhea has a tendency to be watery, voluminous and persistent even when no feeding is given orally. It is usually caused by an external or internal secretagogue (cholera toxin, lactase deficiency). Osmotic diarrhea follows ingestion of a poorly absorbed solute because of an inherent character of the solute (magnesium phosphate, alcohol, sorbitol) or a small bowel defect (lactose in lactase deficiency in brush border). It tends to be watery and acidic with reducing substances. Motility diarrhea is associated with increased (irritable bowel syndrome) or delayed motility (intestinal pseudo obstruction).

- **Acute diarrhea** refers to diarrhea that begins acutely and terminates within a week or so, only a small proportion of cases passing to the second week or even beyond.
- **Chronic diarrhea** refers to diarrhea beyond 2 weeks. The term is best reserved for cases with an obvious malabsorption or an underlying organic disease without obvious malabsorption.

The term persistent diarrhea denotes an episode of acute diarrhea, presumably of infective origin, that lasts for 2 weeks or more. The term, intractable diarrhea of infancy, should be reserved for cases who have onset of protracted diarrhea before the age of 3 months. These infants start as an infective diarrhea, become dehydrated and wasted and have high mortality. They need emergency treatment.

1.2.1 ACUTE DIARRHEA: -

Acute diarrhea, often called acute gastroenteritis, is, in particular, a leading cause of mortality in pediatric practice. This emergency accounts for as high as 75% of our pediatric admissions at certain times during the peak summer and rainy seasons. The incidence and mortality are especially high in infancy, more so in the presence of malnutrition and erratic feeding practices. According to a conservative estimate, almost 500 million children suffer from acute diarrhea annually. Of them, 5 million die every year. In India alone, nearly 1.5 million children die due to acute diarrhea every year [1].

1.2.1.1 **ETIOLOGY: -**

That acute diarrhea is mostly infectious in origin in pediatric practice is borne out by the following points:

1. Magnitude of diarrhea prevalence is directly proportional to sanitary and personal hygiene standards of the community.

2. Acute diarrhea in the community behaves on the same lines as other infectious diseases Infants and children are more frequently and more severely affected than older people, indicating poor immunity in the former.

1.2.1.2 **Etiology of pediatric acute diarrhea**

A. Enteric Infections Bacteria: E. coli, Shigella, Salmonella, Staphylococcus, Cholera vibrio, Yersinia enterocolitica, Campylobacter jejuni, Clostridium difficile, Aeromonas hydrophilia. Vibrio parahaemolyticus, Plesiomonas shigelloses.

Viruses: Rotavirus, Norwalk and allied viruses, enterovirus. Influenza virus, measles virus.

Parasites: Ent. histolytica, Giardia lamblia, Cryptosporidium, H. nana, malaria

Fungi: Candida albicans.

Parental: URI, otitis media, tonsillitis, pneumonia, urinary tract infection.

B. Dietetic/Nutritional: Overfeeding, starvation, food allergy, food poisoning.

C. Drugs: Antibiotics

D. Nonspecific.

1.2.1.3 **Pathogenesis**

The delicate balance in the ecology of the gastrointestinal tract needs to be broken down by any of the two situations in order that diarrhea occurs:

1. The conditions in which the defense weakens, say malnutrition (both primary and secondary), immunologic disorders, etc. so that even commensals, organisms with weak virulence, or opportunist organisms overpower and cause diarrhea.

2. The known pathogens overcome the natural defense.

The pathogenic organisms produce diarrhea by one or more of the following mechanisms:

1. Adhesion to the intestinal mucosal wall, e.g. enteropathogenic E. coli (EPEC) which are further categorized as class I EPEC (showing localized adherence) and class II EPEC (showing diffuse adherence) [1].

2. Elaboration of an exotoxin, (secretory diarrhea) e.g. rotavirus, enterotoxigenic E. coli (ETEC), Vibrio cholera, Aeromonas hydrophilia, Plesiomonas shigelloides, causes excessive

3. Mucosal invasion (exudative diarrhea), e.g. enteroinvasive E. coli (EIEC), Shigella, Salmonella (non-typhi), Cl. Defficile, Campylobacter jejuni, Yersiniaenterocolitica, ent[redacted]hogenic E. coli (EPEC), rotavirus, damage and exudative blood. No effect of fasting. The term, Osmotic diarrhea, denotes diarrhea from concentrated substances (antacids, lactose, w[redacted] are not absorbed from the gut. They pull water from intestinal wall into stools. Such diarrhea subsides on fasting. In order to appreciate the modus operandi of diarrheal dehydration and dyselectrolytemia, it must be noted that diarrheal losses are drawn from extracellular fluid (ECF) compartment constituted by circulating blood, interstitial fluid and secretions. This compartment accounts for the 60% of the body weight of the child. Loss of water from the child's body causes shrinkage in the volume of the ECF compartment. In around one-half cases with ex[redacted]e sodium loss in stools, hyponatremia develops as a result of fall in serum and ECF sodium levels. Since sodium is the major determinant of osmolality, there results fall in osmolality of ECF. Then follows movement of water from ECF to ICF compartment. Further shrinkage of the already shrunk ECF compartment volume becomes inevitable. This is manifested in the form of loss or impairment of skin elasticity. In a small proportion of [redacted] in whom diarrheal dehydration has been treated with fluids containing far too much of sodium, osmotic pressure of ECF becomes high, prompting water in ICF c[redacted]rtment to move to ECF compartment. This is likely to camouflage the existence of severe dehydration which may erroneously be interpreted as mild dehydration. Depletion of ECF c[redacted]rtment leads to reduction in blood volume, causing peripheral circulatory failure and oliguria or anuria. Loss of potassium in stools leads to hypokalemia, causing abdominal distention, hypotonia and ECG changes in the form of ST depression and flat T wave. Loss of bicarbonate in stools leads to acidemia, causing acidotic respiration (Kussmaul breathing) which is characteristically deep and rapid.

1.2.1.4 Clinical Features

The clinical picture varies in mild, moderate and severe cases Mild In mild ca[redacted] onset is usually insidious with 2 to 5 motions which may be loose, green, offensive and contain mucus and milk curds. The volume may be small or large. The attack usually subsides in a day or two without any remarkable constitution[redacted] [redacted]anifestations or dehydrations [1]. Moderate The number of motions is 10 or more and constitutional symptoms like fever, irritability, anorexia and vomiting is usually present. Mild dehydration (3 to 5%) is associated. Severe Here the child passes "too many" loose motions and has severe vomiting to the extent that nothing is retained and the oral intake becomes virtually impracticable. Such cases are most often characterized by sudden rather than gradual onset. They may have marked constitutional symptoms [1]

TABLE 1.1 CLINICAL FEATURES OF VIRAL VS CHRONIC DIARRHEA

Features	Viral diarrhea	Bacterial diarrhea	
		Invasive	Non invasive
Character of motion	watery	Watery or semisolid with mucus but no blood	Semisolid, small amount, very frequent with mucus and blood
Vomiting	Severe	Only slight	Moderate
Pyrexia	Slight	Nil	Moderate to high
Upper respiratory tract infection	Usually present	Nil	Nil
Seizures	Nil	Nil	Occasionally
Toxemia	Nil	Nil	Slight
Stool microscopy	Moderate pus cells	NAD	Moderate pus and red cells

TABLE 1.2 CLINICAL PICTURE IN ISOTONIC, HYPOTONIC AND HYPERTONIC DEHYDRATION

Criteria	Isotonic	HYPERTONIC	HYPOTONIC
Skin color	Gray	Gray	Gray
Temperature	Cold	Cold or hot	Cold
Turgor	Poor	Fair	Very poor
Feel	Dry	Thickened	Clammy(moist)
Mucous membrane	Dry	Parched	Slightly moist
Eyes	Sunken &soft	Sunken	Sunken &soft
Anterior fontanel	Depressed	Depressed	Depressed
Sensorium	Drowsy	Very irritable	Comatose
Pulse	Rapid	Moderately rapid	Rapid
Blood pressure	Low	Moderately low	Very low

1.2.1.5 Clinical Assessment of Diarrheal Dehydration-

Since la[redacted]atory investigations are most often not available, it is advisable to make the best use of one's clinical knowledge in evaluating the grades and type of dehydration.

TABLE 1.3 ASSESSMENT OF DIARRHEAL DEHYDRATION AS PER WHO

Area of clinical observation	Actual observation		
	No dehydration	Some dehydration	severe dehydration
General condition	Well alert	Restless, irritable	Lethargic, Unconscious
Eyes	Normal	Sunken	Sunken
Thirst	Drink normally, not thirsty	Drink eagerly, thirsty	Drinks poorly unable to drink.
Skin pinch	Goes back quickly	Goes back slowly	Goes back very quickly

1.2.1.6 Treatment

(A) Conventional Rehydration Therapy Replacement of the fluids as soon as possible, is the sheet-anchor of management of acute diarrhea.

1. Oral rehydration therapy [redacted]T), as described in details later in this very Chapter, is ideal for mild dehydration and a majority of the children with moderate dehydration. One may use the standard WHO ORS, one of its modifications, or a home- made electrolyte solution (HES). There are distinct advantages in using a cereal-based solution such as rice water electrolyte solution (RWESD), especially since it has a better tolerance and provides greater energy. Each motion must be followed by replacement with an equal amount of ORS. Breastfeeding must not be discontinued. In fact, it potentiates the usefulness of ORT

Intravenous fluid therapy is indicated in cases with severe dehydration [1].

(B) Symptomatic Control

In the past, nonspecific antidiarrheal agents like codeine, morphia, tincture opium, charcoal, chalk, anticholinergic drugs, products of hydroscopic bulk (psyllium seed or plantago ovtum), kaolin, bismuth, pectin, diphenoxylate hydrochloride and loperamide hydrochloride have been used. These drugs are either not quite effective or their use is accompanied by unpleasant/untoward side-effects. These are no longer recommended. The role of prostaglandin inhibitors and antisecretory agents such as aspirin, though theoretically significant, needs detailed evaluation in the therapy.

(C) Diet

Prolonged "starvation" damages rather than helps and should be discouraged. Even hypocaloric oral therapy during an episode of diarrhea and vomiting may lead to severe malnutrition. Lack of attention to nutrition during diarrhea appears to be the largest contributor factor to overwhelming problem of malnutrition in the Indian subcontinent. Banana, apple pulp, yoghurt, curd, potatoes, rice, wheat, etc. should be given as soon as possible. Foods rich in fats or sugar, including juices and soft drinks, should be avoided.

Current recommendations on nutritional management of acute diarrhea are as follows:

- Since most nutrients are well-absorbed during diarrhea and since diarrhea predisposes to malnutrition, it is safe and desirable to continue breastfeeding as also other feeding during a diarrheal episode. That rest to gut promotes early recovery is no longer held true. It has no physiologic basis at all.
- Optimally energy-dense foods with minimal bulk, given in small quantities every 2 to 3 hours, promote better nutrition.
- Since staple foods do not provide optimal calories per unit weight, these are best enriched with richer sources of energy like fats and oils, e.g. Khichri with oil, rice with milk or curd, mashed banana with milk or curd, mashed potato with oil etc.
- Foods with high fiber content (coarse fruits and vegetables) as also soft drinks and juices with very high sugar content may be avoided during an acute diarrheal episode.
- In artificially-fed infants, milk should preferably be given undiluted during all phases of acute
diarrhea. If the infant is over 4 months, milk cereal mixture (say dalia-sago, rice-milk) is strongly recommended [1].
- Transient lactose intolerance, which is frequent in acute diarrheal disease, does not warrant

lactose free milk unless it persists beyond 8 to 10 days and is accompanied by progressive weight loss.

• During convalescence from acute diarrhea, dietary intake should be enhanced by at least 25% of normal to make up for the losses during illness and to promote rapid weight gain until the child attains normal nutritional status. Finally, it is most appropriate to re-emphas[redacted] WHO/UNICEF slogan that the full package for diarrhea therapy in a vast majority of children is ORS and continued feeding.

(D) Ancillary Measures

These include control of vomiting by sips of ORS, a mild [redacted]etic or stomach wash, and treatment of any other accompanying problem. If IV drip is to be prolonged, vitamins should be added to the infusion.

It is advisable [redacted]tore the normal intestinal flora (lactobacilli) which may be destroyed by the disease or by antibiotic therapy by administering a preparation containing lactobacillus strains. Abdominal distention, if mild and with normal bowel sounds, [redacted]nt no intervention. Paralytic ileus, manifested by gross abdominal distention and poor or absent bowel sounds, is an indication for temporary withdrawal of oral feeds, intermittent nasogastric suction and administration of potassium chloride with a parenteral fluid. The existence of septicemia or en[redacted]olitis should be seriously considered and treated energetically. Zine, 20 mg/day (O), in every child with diarrhea, for 2 weeks is recommended by the IAP.

1.2.1.7 Prognosis

i. Age: Mortality is higher in newborns and infants than in older children.

ii. Nutritional status: Diarrhea in malnourished children carries poor prognosis. Even mild to moderate diarrhea in such subjects may cause almost irreversible metabolic alterations, causing death. In one investigation, while the mortality in well-nourished patients was 4.3%, it was 22% in those suffering from mara[redacted]

iii. Causative organism and severity of illness: E. coli resistant to most available antibiotics and Shigella cause very severe illness.

iv. Associated illness/complications: Presence of profound dehydration, electrolyte imbalance or bronchopneumonia definitely has adverse effect on the outcome.

v. Management: Promptness and adequacy of treatment also have great bearing on the ultimate outcome [1].

1.2.1.8 Prevention

i. Improvement in the nutritional status of the children. Malnutrition predisposes to diarrhea which further aggravates the state of poor nutrition.

ii. Improvement in community's water supply, sanitation and hygiene. Mothers must ensure proper hand washing before serving, preparing or eating food, using clean (potable), preferably boiled or filtered, drinking water, protecting food from contamination by flies, cockroaches and dirt, washing fruits and vegetables before use, and proper disposal of excreta

iii. Breast (biological) feeding should be encouraged Those who are bound to stick to artificial feeding should learn the hygienic preparation of the formula and care of bottle, teats, etc.

iv. Mothers must be taught when to consult the doctor in case of diarrhea.

v. Standardized simple method of administering intravenous fluids should be available not only in the cities but in rural areas as well.

vi. Easy availability of "take home" ORS sachets

vii. Rotavirus vaccine an oral tetravalent rotavirus vaccine against all four major serotypes incorporating RRV and reabsorbent viruses (human serotype Gl, G2 and G4) in a single vaccine is now available. ETEC vaccine and cholera vaccine are available whereas a live shigella vaccine is around the corner.

1.2.1.9 Complications/Sequelae

• Dehydration with its widespread complications, including renal shutdown, paralytic ileus, thromboembolism, seizures.

• Superadded infections including thrombophlebitis at the site of cutdown.

• Overhydration and CCF.

• Malnutrition.

• Hypoglycemia.

• Syndrome of inappropriate secretion of antidiuretic hormone (ADH).

• Carbohydrate intolerance and chronic diarrhea.

• Subdural collection of fluid/blood that may possibly cause mental retardation in later life.

• Consumptive coagulopathy.

• Toxic megacolon [1].

1.3 CHRONIC DIARRHEA

1.3.1 Definition

Chronic diarrhea is defined as "diarrhea of at least 2 weeks duration or 3 attacks of diarrhea during the last 3 months", usually due to obvious malabsorption or an organic or other cause countries and is responsible for considerable ill health and morbidity

without obvious malabsorption. It is a common pediatric problem in tropical

1.3.2 Evaluation Protocol

Roughly diagnostic evaluation of the child with chronic diarrhea should be step-by-step rather than by a large number of investigations at a time. The individual merits of each case and the proper application of knowledge and experience of the attending pediatrician contribute to deciding the necessary investigations.

1.3.2.1 Four phases of evaluation of the child with chronic diarrhea/malabsorption

Phase I: History and physical examination with special reference to onset of diarrhea and its relationship with various factors (excessive carbonated drinks/ fruit juices, supplementary milk feeds, cereals), specific amount of fluids ingested/day, nutritional status, etc. Meticulous stool examination (ova and cysts, pH, reducing substances, fat globules) Stool culture Stool for Cl. difficile toxic Blood studies (CBC, ESR, electrolytes, BUN, creatinine

Phase II: Fat balance studies for daily stool fat or steatocrit D-xylose test Sweat chloride test Stool osmolality and electrolytes, phenolphthalein, magnesium sulfate, phosphate Breath H2 tests.

Phase III: Barium meal/enema to exclude anatomic defects small intestinal biopsy/colonic biopsy by endoscopic studies Sigmoidoscopy/colonoscopy.

Phase IV: Hormonal studies Neurotransmitter studies (vasoactive intestinal poly- peptide, gastrin, secretin, 5-hydroxyindoleacetic assays) [1].

1.3.3 Pathophysiologic Mechanisms

Osmotic diarrhea results from presence of malabsorption of water-soluble nutrients (lactose intolerance) and excessive intake of carbonated fluids or nonabsorbable solutes (sorbitol, lactulose, magnesium hydroxide) which cause an osmotic load in the colon. It shows good response to simple fasting. Secretory diarrhea results from activation of intracellular mediators like cyclic adenosine monophosphate (cholera, heat-labile E. coli, Shigella, Salmonella. jejuni, Ps. aeruginosa, hormones like vasoactive intestinal peptide, gastrin, secretin, anion surfactants like bile acids and ri[redacted]ic acid), cyclic guanosine monophosphate (heat stable E. coli, Y. enterocolitica and intracellular calcium (Cl. difficile, acetylcholine, serotonin, brady-kinin).

Mutation defects in apical membrane (ion) transport proteins such as in chloride-bicarbonate exchange and sodium-bile acid transporter result in secretory diarrhea and failure to thrive (FTT) at birth. Reduction in anatomic surface area in such conditions as short bowel syndrome following surgical resection in necrotizing enterocolitis, valvulitis or atresia. Alteration in intestinal motility in such conditions as malnutrition, diabetes mellitus, intestinal pseudo-obstruction syndromes and scleroderma. Here, diarrhea is of secretory type.

1.3.4 Etiologic Considerations

A large number of conditions, involving intraluminal factors, mucosal factors, or both, can cause chronic diarrhea. Nevertheless, the scene is dominated by a few conditions.

1. Is the child consuming excessive amounts of carbonated drinks or fruit juices (over 150 ml/kg/24 hours) and yet has normal growth and height param[redacted] (nonspecific chronic diarrhea)? The problem usually resolves following reduction in fluids (under 90 ml/kg/24 hours) [1].

1.4 MALNUTRITION

An imbalance between nutrient requirement and intake, resulting in cumulative deficits of energy, protein, or micronutri[redacted]hat may negatively affect growth, development and other relevant outcomes [2]. Based on its etiology, malnutrition is either illness related (one or more diseases or injuries directly result in nutrient imbalance) or caused by environmental/behavioral factors associated with decreased nutrient intake and/or delivery. Primary acute malnutrition in children is the result of inadequate food supply caused by socioeconomic, political, and environmental factors, and it is most commonly seen in low- and middle-income countries [3,4].

Responsible factors include:

- household food insecurity
- poverty.
- poor nutrition of pregnant women,
- intra uterine growth restriction,
- low birth weight,
- poor breastfeeding and inadequate complementary feeding,

- ➢ frequent infectious
- ➢ illnesses,
- ➢ poor quality of water, hygiene,

Primary acute malnutrition is mostly social rather than biomedical in origin, but it is also multifactorial. For example, poor water quality, sanitation and hygiene practices are increasingly believed to be the cause of the condition called "environmental enteropathy" that contributes to acute malnutrition in childhood [5]. The repetitive exposure to pathogens in the environment causes small intestinal bacterial colonization, with accumulation of inflammatory cells in the small intestinal mucosa, damage of intestinal villi, and, consequently, malabsorption of nutrients, which results in malnutrition. Secondary acute malnutrition is usually due to [redacted]mal nutrient loss, increased energy expenditure, or decreased food intake, frequently in the context of underlying, mostly chronic, diseases like cystic fibrosis, chronic renal failure, chronic liver diseases, childhood malignancies, congenital heart disease, and neuromuscular disease [3,4].

1.4.1 Pathoph[redacted]gy

Inadequate energy intake leads to various physiologic adaptations, including growth restriction, loss of fat, muscle, and visceral mass, reduced basal metabolic rate, and reduced total energy expenditure [3, 4, 5].

The biochemical changes in acute malnutrition involve metabolic, hormonal, and glucoregulatory mechanisms. The main hormones affected are the thyroid hormones, insulin, and the growth hormone (GH). Changes include reduced leve[redacted]tri-iodothyroxine (T3), insulin, insulin-like growth factor-1 (IGF-1) and raised levels of GH and cortisol [3]. Glucose levels are often initially low, with depletion of glycogen stores. In the early phase there is rapid gluconeogenesis with resultant loss of skeletal muscle caused by use of amino acids, pyruvate and lactate. Later there is the protein c[redacted]vation phase, with fat mobilization leading to lipolysis and ketogenesis [6,7,8]. Major electrolyte changes including sodium retention and intracellular potassium depletion can be explained by decreased activity of the glycoside-sensitive energy-dependent sodium pump to increased permeability of cell membranes in kwashiorkor [8]. Organ systems are variably impaired in acute malnutrition [3, 8]. Cellular immunity is affected because of atrophy of the thymus, lymph nodes, and tonsils.

There are reduced cluster of differentiation (CD) 4 with normal CD8-T lymphocytes, loss of delayed hypersensitivity, impaired phagocytosis, and reduced secretory immunoglobulin A.

Consequently, the [redacted]ptibility to invasive infections (urinary, gastrointestinal infections, septicemia, etc.) is increased [8,9] Villous atrophy with resultant loss of disaccharidases, crypt hypoplasia, and altered intestinal permeability results in malabsorption. Other common aspects are ba[redacted]l overgrowth and pancreatic atrophy resulting in fat malabsorption; fatty infiltration of the liver is also common [3]. Drug me[redacted]sm may be decreased due to decreased plasma albumin and decreased fractions of the glycoprotein responsible for binding drugs [10]. Cardiac myofibrils are thinned with impaired contractility. Cardiac output is reduced proportionate to weight loss. Bradycardia and hypotension are also common in severe cases [3, 9]. The combination of bradycardia, impaired cardiac contractility, and electrolyte imbalances predisposes to arrhythmias. Reduced thoracic muscle mass, decreased metabolic rate, and electrolyte imbalances (hypokalemia and hypophosphatemia) may result in decreased minute ventilation and impaired ventilatory re[redacted]e to hypoxia [3, 9,11]. Acute malnutrition has been recognized as causing reduction in the numbers of neurons, synapses, dendritic arborizations, and myelinations, all of which resulting in decreased brain size. The cerebral cortex is thinned and brain growth slowed. Delays in global function, motor function, and memory have been associated with malnutrition [12]. The effects on the developing brain may be irreversible
after the age of 3–4 years [4].

Fig NO. 1.3: Malnourished child

1.4.2 Clinical Syndromes

Acute malnutrition pertains to a group of linked disorders that includes kwashiorkor, marasmus, and intermediate states of marasmic kwashiorkor. They are distinguished based on clinical findings, with the primary distinction between kwashiorkor and marasmus being the presence of edema [redacted]ashiorkor [9].

1.4.2.1 Marasmus

The term "marasmus" is inferred from the Greek word "marasmus", correlating to wasting or withering. Marasmus is the most frequent syndrome of acute malnutrition [3]. It is due to inadequate energy intake over a period of [redacted]s to years. It results from the body's physiologic adaptive response to starvation in response to severe deprivation of energy and all nutrients, and is characterized by wasting of body tissues, particularly muscles and subcutaneous fat, and is usually a result of severe restrictions in energy [redacted]e. Children younger than five years are the most commonly involved because of their increased caloric requirements and increased susceptibility to infections [8]. These children appear emaciated, weak a[redacted]thargic, and have associated bradycardia, hypotension, and hypothermia. Their skin is xerotic, wrinkled, and loose because of the loss of subcutan[redacted] fat, but is not characterized by any specific dermatosis Muscle wasting often starts in the axilla and groin (grade I), then thighs and buttocks (grade II), followed by chest and abdomen (grade III), and finally the facial muscles (grade IV), which are metabolically less active. In severe cases, the loss of buccal fat pads gives the children an aged facial aspect. Severely affected children are often apathetic but become irritable and difficult to console [3].

1.4.2.2 Kwashiorkor

The term "kwashiorkor" derives from the Kwa language of Ghana and its meaning is equivalent to "the sickness of the weaning [8]. "Kwashiorkor is thought to be [redacted] result of inadequate protein but reasonably normal caloric intake. It was first reported in children with maize diets (these children have been called "sugar babies", as the[redacted] is typically low in protein but high in carbohydrate) [3, 8]. Kwashiorkor is frequent in developing countries and mainly involves older in[redacted] and young children. It mostly occurs in areas of famine or with limited food supply, and particularly in those countries where the diet consists mainly of corn, rice and beans [13]. Kwashiorkor represents a maladaptive response to starvation. Edema is the distinguishing characteristic of kwashiorkor, which does not exist in marasmus [15], and usually results from a combination of low serum albumin, increased cortisol, and inability to activate the antidiuretic hormone. It usually starts as pedal edema (grade I), then facial edema (grade II), paraspinal and chest edema (grade III) up to the association with ascites (grade IV). Besides edema, clinical features are almost normal weight for age, dermatoses, hypopigmented hair, distended abdomen, and hepatomegaly. Hair is usually dry, sparse, brittle, and depigmented, appearing reddish yellow. Cutaneous manifestations are characteristic and progress over days from dry atrophic skin with confluent areas of hyperkeratosis and hyperpigmentation, which then splits when stretched, resulting in erosions and underlying erythematous skin. Various skin changes in children with kwashiorkor include shiny, varnished-looking skin (64%), dark erythematous pigmented macules (48%), xerotic crazy paving skin (28%), residual hypopigmentation (18%), and hyperpigmentation and erythema (11%) [3].

1.4.2.3 Marasmic Kwashiorkor

Marasmic kwashiorkor is represented by mixed features of both marasmus and kwashiorkor. Characteristically, children with Marasmic kwashiorkor have concurrent gross wasting and edema. They usually have mild cutaneous and hair manifestations and an enlarged palpable fatty liver.

1.4.3 Assessment of Protein Energy Malnutrition

Nutritional assessment can be approached through four basic procedures/methods: history, clinical examination, anthropometric measurements, and laboratory test [15].

1.4.3.1 History:

Malnutrition should be suspected in any child whose dietary history reveals deficiency in the quantity or quality of food intake. Children at risk of malnutrition may have a history of one or more of the following: premature birth or low birth weight; lack of breastfeeding in poor social circumstances; twin birth prolonged or recurrent diarrhoea, pneumonia, tuberculosis or other infectious diseases; affectation or infection by HIV; cancer or chronic disease [16].

1.4.3.2 Clinical Signs of Malnutrition

The following clinical signs are often recognized

- **(A) Hair changes** includes silkiness, curliness or straightness; lightly colored to shades of brown/red; lack of luster; sparsely distributed to alopecia thin and easily pluckable [15,16].
- **(B) Dermatosis:** This is commoner in the oedematous PEM and manifest as hypo or hyperpigmentation, rough patches (hyperkeratosis), shedding of the skin in scales and

sheets, cracks and fissures that may ooze plasma. Extensive skin desquamation may present appearances like second degree burns [15,16].

(C) **Stomatitis:** This typically manifests as angular stomatitis and glossitis with loss of papilla [15,16] is, may be related to deficiency of micronutrients like iron or vitamin B2and are often complicated by infection [17].

(D) **Visible Signs of Wasting:** Wasting is most visible in the following areas: chest as prominent ribs, arms and thighs as loose skin and flabby muscle, the back as prominent scapula and spine, and the buttocks as loss of fat and muscles with the overlying skin hanging loose in a typical "baggy pants" fashion.

(E) **Peripheral Oedema:** The presence of bilateral pedal oedema in a child should alert health practitioners on the possible diagnosis of edematous malnutrition, particularly in the presence of other physical signs of malnutrition.

1.4.3.3 Anthropometric Indicators of Malnutrition

Anthropometry is the determination of nutritional status by physical measurements and comparing them to relevant reference charts such as the WHO. weight-for-height reference chart.

(A) **Height-for-age** is used to assess linear growth. Deficit indicates long-term, cumulative nutritional inadequacies [18]. Children whose height-for-age indices fall below 90% of the median value (< –2SD) of the WHO reference value are classified as stunted with those below 185% (<–3SD) being severe. Because deficit in height results from a long-term process, stunting denotes chronic mal-nutrition. Length is measured for children less than of age 2 years whilst standing height is done for others.

(B) **Weight-for-age** index has traditionally been used in defining malnutrition with children whose measurements fall below 80% of the median value (< – 2 SD) being classified as malnourished. Because low weight-for-age may be due to low height for-age (stunting), low weight-for-height(wasting), or both (global malnutrition), weight-for-age is not currently are commended measurement to define acute malnutrition [18].

(C) **Weight-for-height** is the most objective way of assessing for recent nutritional inadequacies. Its use carries the advantage of requiring no knowledge of age which is often difficult to obtain in most developing countries. Children whose measurements fall below 80% of the median value (< – 2 SD) of the WHO reference value are classified as wasted with those below 70% (< –3SD) being severe. Weight-for-height is the current recommended measurement for defining acute malnutrition.

(D) **Mid-upper arm circumference (MUAC)** has been used as alternative index for assessing the nutritional status of children especially where th[redacted]llection of height and weight measurements may be difficult as, for example, in emergency situations like refugee crises and famines [18].

Fig NO. 1.4: Mid upper arm circumferenece measuring

1.4.3.4 Laboratory findings Low serum prealbumin and albumin, depressed blood urea, profoundly low serum cholesterol, and low hemoglobin and transfer [15].

1.4.4 MANAGEMENT OF MALNUTRITION &DIARRHEA

Initially the major emphasis of treatment of acute diarrhoea in children is the prevention and treatment of dehydration, electrolyte abnormalities and comorbid conditions (e.g. pneumonia, bacteremia). Nutritional management is, however, not delayed and is an integral part of their management. Objectives are to prevent weight loss, to encourage catch-up growth during recovery, to shorten the duration and to decrease the impact of the diarrhoea on the child's health. Feeds are only interrupted for a few hours to allow correction of severe dehydration and shock [19]. There is no indication for "regrading" of feeds in infants[redacted]ing diluted or low volumes of feeds aggravates the weight loss that follows an episode of diarrhoea and does not reduce the duration or severity of the diarrhoea significantly. Breastfeeding should not be interrupted; it shortens the duration of the diarrhoea and improves energy intake. Initially children continue with their normal feeds and specialized formulae are indicated for those with malabsorption. Lactose free formulae are given to children with suspected lactose intolerance; this occurs more frequently in malnourished children and those whose diarrhoea persists longer than 3–4 days. Hydrolyzed infant formulae are used for those with suspected protein sensitive enteropathy and may be of value in some children with severe malnutrition or HIV infection [20].

1.4.4.1 Zinc

Zinc, a cofactor for many enzymes and required for cell division, has a number of potentially beneficial effects on the intestine. It reduces intestinal permeability in malnourished children with diarrhoea [21,22] and, in animal models, tight junction morphology is improved and paracellular permeability decreases after zinc supplementation [23]. Clinical benefits of zinc supplementation include reduction of the severity and duration of diarrhoea in children at high risk of zinc deficiency. A recent meta-analysis [24] of randomized, placebo-controlled trials of zinc supplementation in children with diarrhoea found that zinc supplementation reduced the duration of acute diarrhoea by 15% and the duration of persistent diarrhoea by 15,5%. Duration of diarrhoea was also reduced by 18,8% and 12,5% for acute and persistent diarrhoea respectively. Weight and length gain are also improved in children after zinc supplementation. A more recent South African study [25], however, did not find the same benefit in children with diarrhoea.

1.4.4.2 Vitamin A

Studies from developing countries have found that vitamin A supplementation reduces the number of episodes of severe diarrhoea and diarrhoeal disease deaths in children. However, there is less evidence that vitamin supplementation will benefit children once they have diarrhoea. Supplementation with vitamin A also reduces intestinal permeability in children in developing countries supporting the role of vitamin A in maintaining the intestinal barrier Function [26,27].

1.4.4.3 Novel Agents

The role of glutamine and its derivatives and arginine in the management of acute infectious diarrhoea are not well established. These agents promote mucosal recovery and may accelerate recovery from diarrhoea.

Malnutrition is a major problem globally. Mal[redacted]hed children have more severe diarrhea, which last longer. Malnutri[redacted]nteracts with diarrhea in a vicious circle leading to high morbidity and mortality. Both malnutrition and electrolyte disturbances are considered to be risk factors for death among children with diarrhea.

- ➢ The present study plans to determine the serum electrolyte status in malnourished children with diarrhea, so[redacted] serum electrolyte disturbances could be managed immediately to reduce the risk of mortality and morbidity. As it is very important to determine and then immediately correct these serum electrolytes changes to prevent the serious life-threatening conditions and decrease the mortality due to diarrhea and malnutrition.

Aim: -

5

- To evaluate the serum electrolyte abnormalities in malnourished children with diarrhea.

Objectives: -

- To assess the serum electrolyte changes in malnourished children with diarrhea.
- To assess the grade of malnutrition.
- To counsel the guardian of malnourished children with diarrhea.

ETHICAL CONSIDERATIONS

One of the major characteristics of modern society is a pronounced interest in ethical questions. Medicine, especially research on humans, is at the top of the list. The Helsinki Declaration, which was established in 1964, (Sixth revision in 2000) is considered as the gold standard for research ethics to provide a universal set of principles and to direct the ethical conduct of medical research. The first principle of ICH (taken from WHO GCP in 1995), states "Clinical studies should be conducted in accordance with the ethical principles that should be consistent with Good Clinical Practice (GCP) and regulatory requirements". The most important ethical aspect of the clinical study is Informed Consent Document (ICD). The international convent accepted by the United Nations Assembly in 1966 stresses that "no one shall be subjected without his fee. Consent to medical or scientific experimentation."

5.1 INFORMED CONSENT DOCUMENT (ICD)

It is of the utmost importance that human volunteers participating in a scientific research project, for instance a clinical study, should understand all details of the planned experiment. The ICD should contain a statement indicating that the study involves research, should describe purpose of research, should contain description of procedures to be followed, expected duration of the volunteer's participation and nature of random assignments. Moreover, ICD should also contain a description of foreseeable risks or discomforts and disclosure of appropriate alternative procedures of treatment that may be advantageous to volunteers. In addition, ICD should include a statement describing the extent to which confidentiality and privacy (of records identifying the volunteers) would be maintained, a statement that participation is voluntary and the refusal to participate involves no penalty or loss. On the basis of these regulations, an informed consent document was prepared for this study (Annexure-II) and properly used.

5.2 STUDY SITE

DEPARTMENT OF PEDIATRIC DEPARTMENT, RAJIV GANDHI INSTITUTE OF MEDICAL SCIENCES (RIMS), KADAPA

The study was conducted in pediatric department in a tertiary care teaching hospital. RIMS Hospital is one of the state governments runner hospitals, managed by senior professionals, highly experienced specialist of departments like general medicine, gastroenterology, psychiatry etc. RIMS hospital also operates all surgical cases especially cardiology, neurology, nephrology, orthopedic.

5.3 Research methodology: -

- **Study design: -** A hospital based prospective study.
- **Study site: -** Pediatric department of government general Hospital (RIMS), Kadapa
- **Study duration: -** six months
- **Sample size:** - 60 patients
- **Study population: -** pediatric patients (6 months to 5 years)

5.4 Patient enrollment: -

5.5 Inclusion criteria-

- 1**.** Malnourished children with diarrhea having age between 6 months to 5 years
- 2.Patients who are willing to participate in the study.

5.6 Exclusion criteria: -

- 1. Malnourished children under 6 months or older than 5 years
- 2. Patients who are not willing to participate in the study
- 3. Children with any congenital or acquired disorder affecting growth

5.8. SOURCE OF DATA

All necessary and relevant information was collected on designed patient data collection form (Annexure-I) which contains demographic details, provisional diagnosis confirmatory diagnosis, radiographic data, social habitats, isolated organism, sensitivity & resistance to various antibiotics, treatment before culture report & after culture report of specimens of patients with CAP in IP general medicine department.

5.9 MATERIALS

1) Patient data collection form (Annexure-I): A well-structured patient data collection sheet was prepared and in which patient details will be recorded.

2) Patient informed consent document (Annexure-II): The details of the patient and laboratory parameters were collected after inform consent taken from the patient.

5.10 Plan of work: -

Literature review on study topic

Preparation of protocol

Submitting for human ethical committee review and approved

Designing of data collection form and informed consent form

Subject recruitment according to eligibility criteria

Monitoring and collection of the data

Interpretation of data to understand the frequency of serum electrolyte abnormalities

Counsel the patient parent (or) guardians

Interpretation of result and preparation of the reports

Communicate the result through publication.

DISTRIBUTION BASED ON GRADE OF MALNUTRITION

TABLE 6.1 Grade wise distribution

GRADE OF MALNUTRITION	FREQUENCY	PERCENTAGE
NORMAL	5	8.33%
GRADE 1	16	26.66%
GRADE 2	22	36.66%
GRADE 3	15	25.00%
GRADE 4	2	3.33%

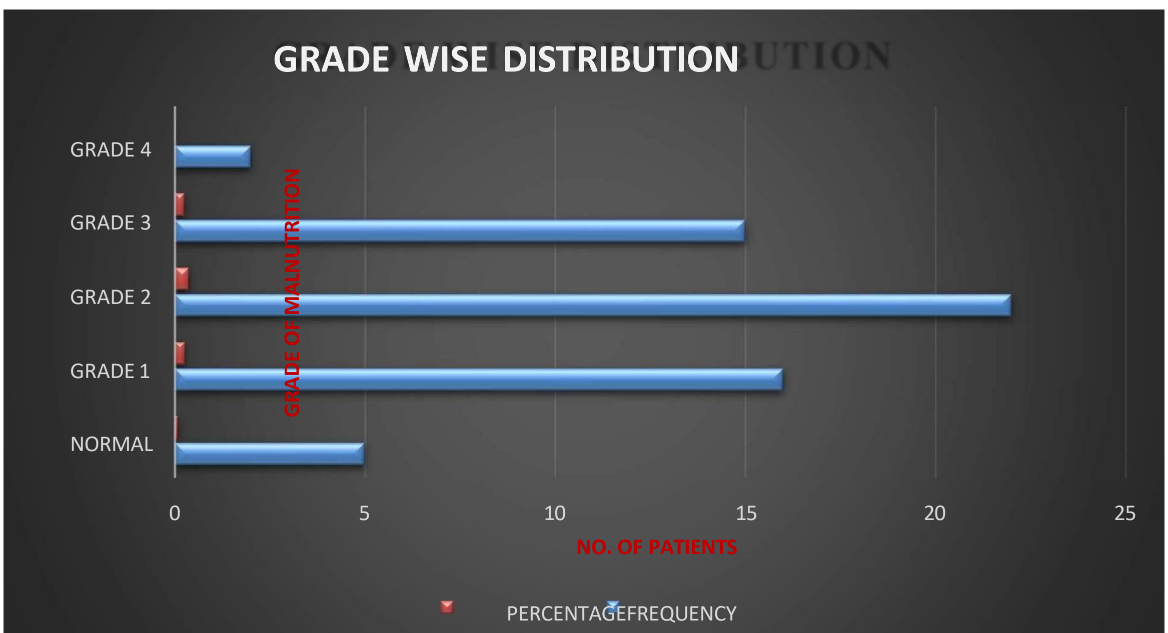

FIGURE .6.1: Grade wise distribution

In this study, 16(26.66%), 22(36.66%), 15(25%) and 2(3.33%) were of Grade 1, Grade 2, Grade 3 and Grade 4. And majority of subjects suffer from grade 2 malnutrition. At this level of malnutrition, the child's weight is only 61 to 70%of the normal child's weight.

6.1 : DISTRIBUTION BASED ON SERUM ELECTROLYTE DISTURBANCES

TABLE 6.2 Distribution of serum electrolytes

SERUM ELECTROLYTES	LOW	NORMAL	HIGH
SODIUM	6	54	0
POTTASSIUM	7	53	0
CHLORIDES	16	39	5

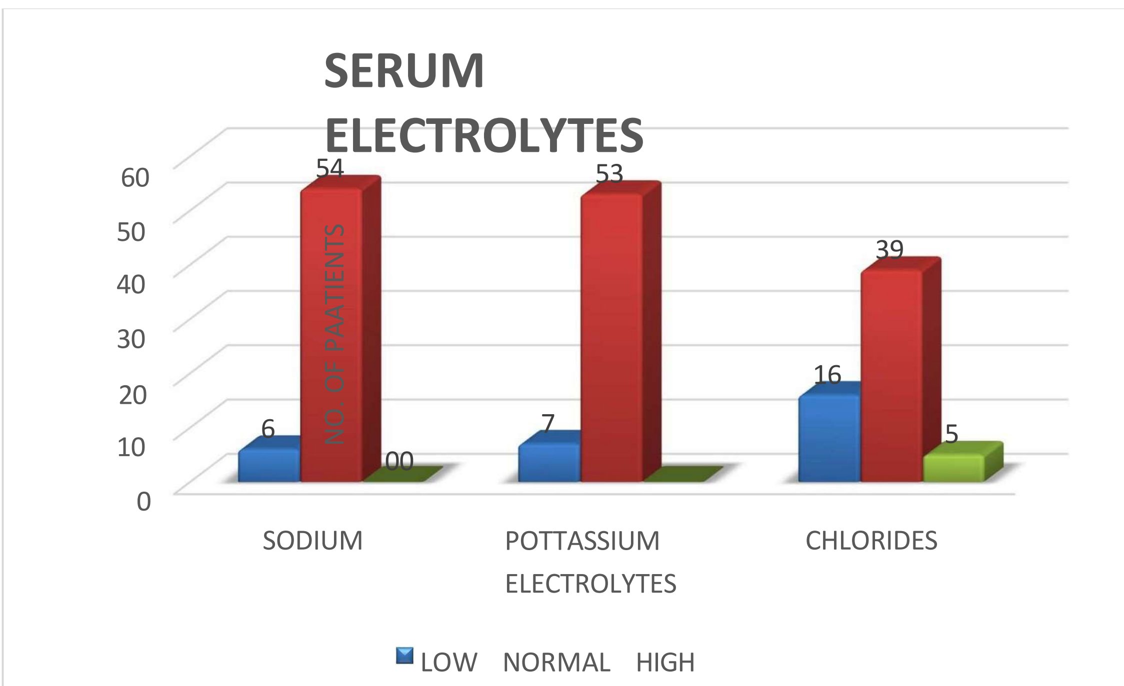

FIGURE. 6.2: Serum electrolyte distribution

Electrolyte disturbances in serum shows normal ranges of sodium, potassium and chlorides in most of the patients.

6.2 : DISTRIBUTION BASED ON GENDER

TABLE 6.3 DISTRIBUTION BASED ON GENDER

GENDER 1	FREQUENCY
MALE CHILD	34
FEMALE CHILD	26

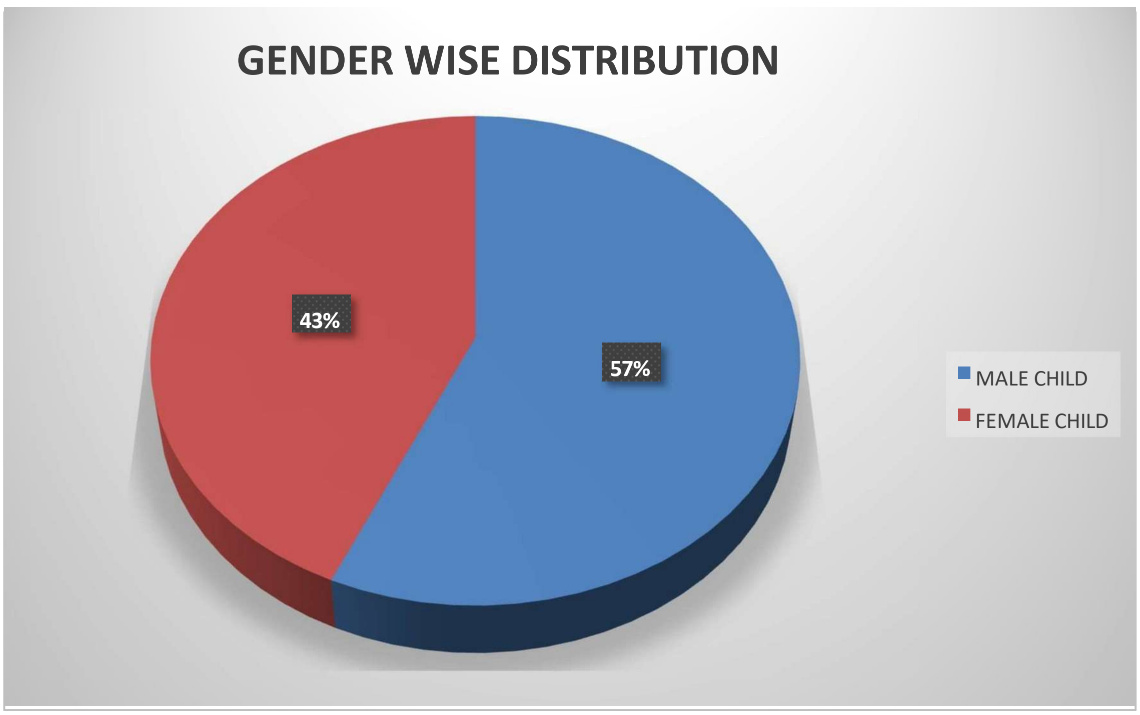

FIGURE 6.3: Grade wise distribution

Among 60 subjects,34(57%) were male children and 26(43%) were female children. In comparison to female subjects, the study population was dominated by male child.

6.3 : DISTRIBUTION BASED ON SEVERITY

TABLE 6.4 SEVERITY

SEVERITY	FEMALES	MALES
MODERATE ACUTE MALNUTRITION	14	21
SEVERE ACUTE MALNUTRITION	12	13

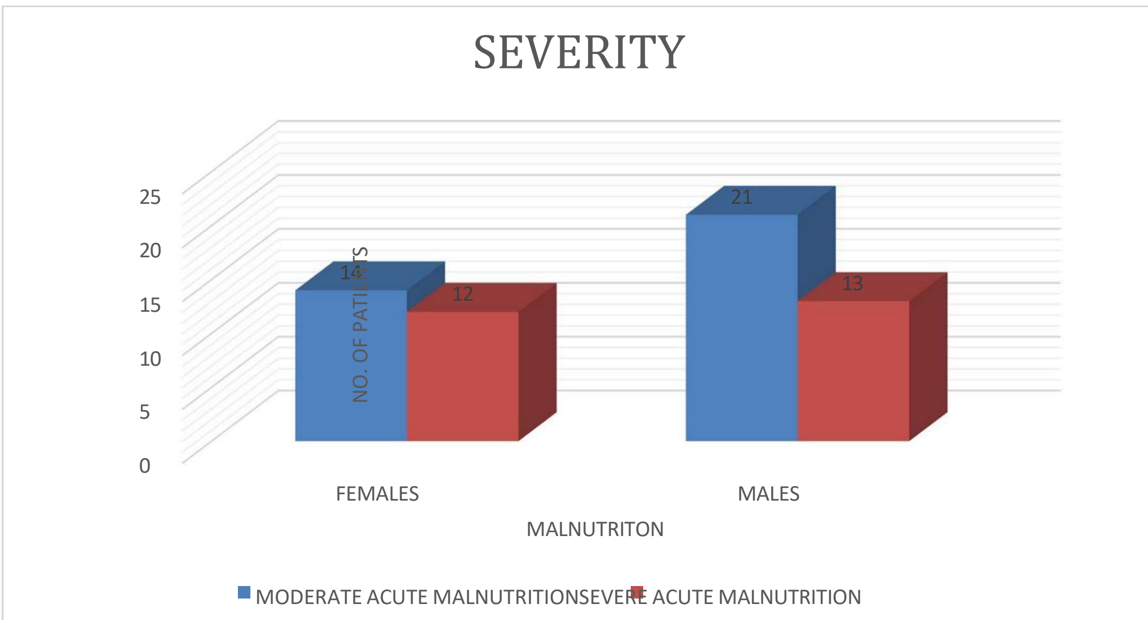

FIGURE 6.4: Distribution based on severity

There were 35 children in the research who were moderately malnourished and 25 who were severely malnourished. The majority of participants had Moderate acute malnutrition.

6.4 : DISTRIBUTION BASED ON MUAC

TABLE 6.5 DISTRIBUTION BASED ON MUAC

MUAC	MALES	FEMALES
<11.5	5	6
11.5-13.5	21	16
>13.5	8	4

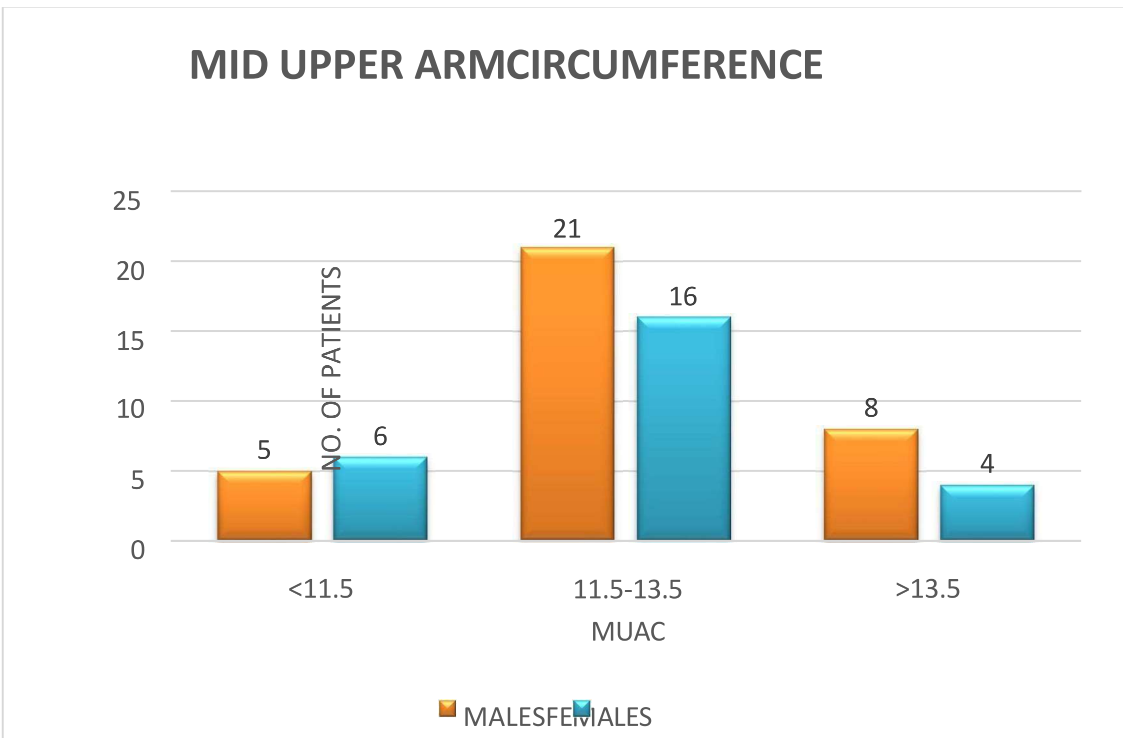

FIGURE.6.5: Based on MUAC

Of the 60 patients, the majority fell into the 11.5 to 13.5 category which indicates severe malnutrition.

6.5 : DISTRIBUTION BASED ON SODIUM LEVELS

TABLE 6.6 DISTRIBUTION BASED ON SODIUM LEVELS

Sodium	N0. of children
Hyponatremia	6
Hypernatremia	0

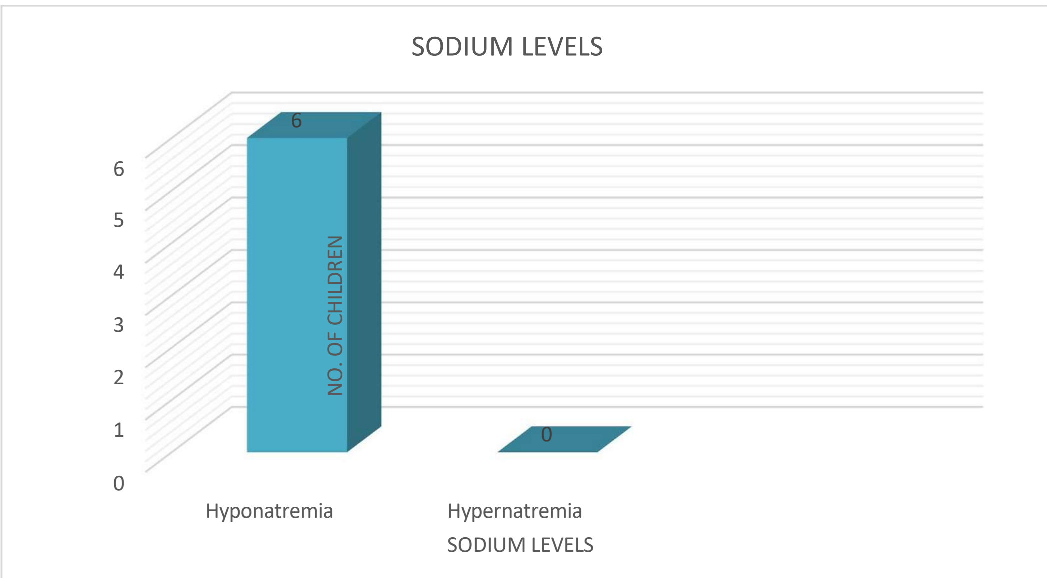

FIGURE.6.6: Distribution based on sodium levels

In this investigation, we discovered that 6 out of 60 patients had hyponatremia.

6.6 : DISTRIBUTION BASED ON POTASSIUM LEVELS

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TABLE 6.7 DISTRIBUTION BASED ON POTASSIUM LEVELS

Potassium	No. of children
Hypokalaemia	7
Hyperkalaemia	0

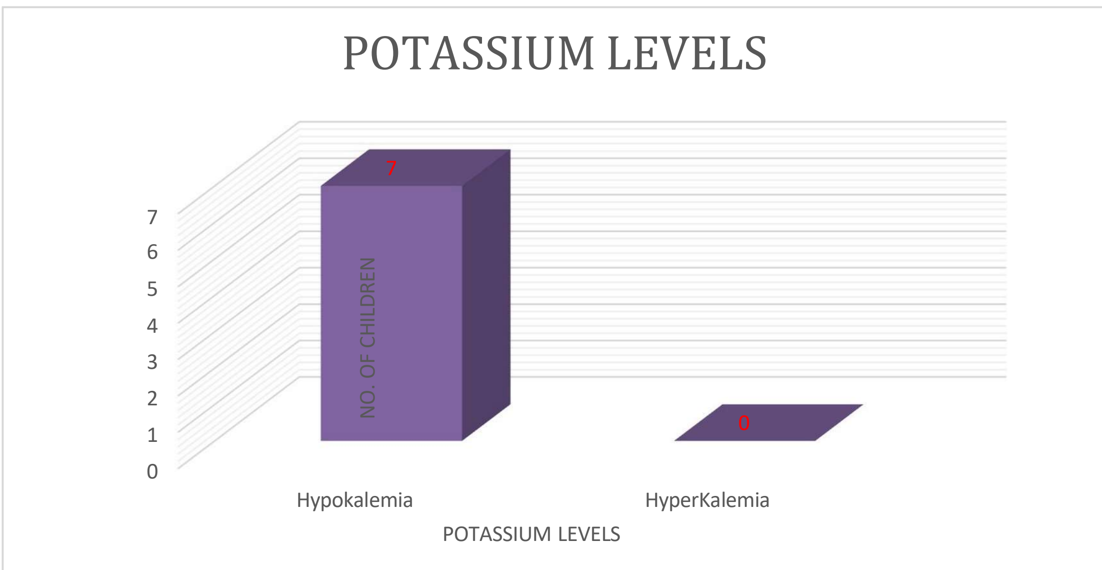

FIGURE.6.7: Distribution based on potassium levels

Out of 60 patients we determined that 7 developed with hypokalaemia.

6.7 : DISTRIBUTION BASED ON CHLORIDE LEVELS

TABLE 6.8 DISTRIBUTION BASED ON CHLORIDE LEVELS

Chlorides	No. of children
Hypochloraemia	16
Hyperchloremia	5

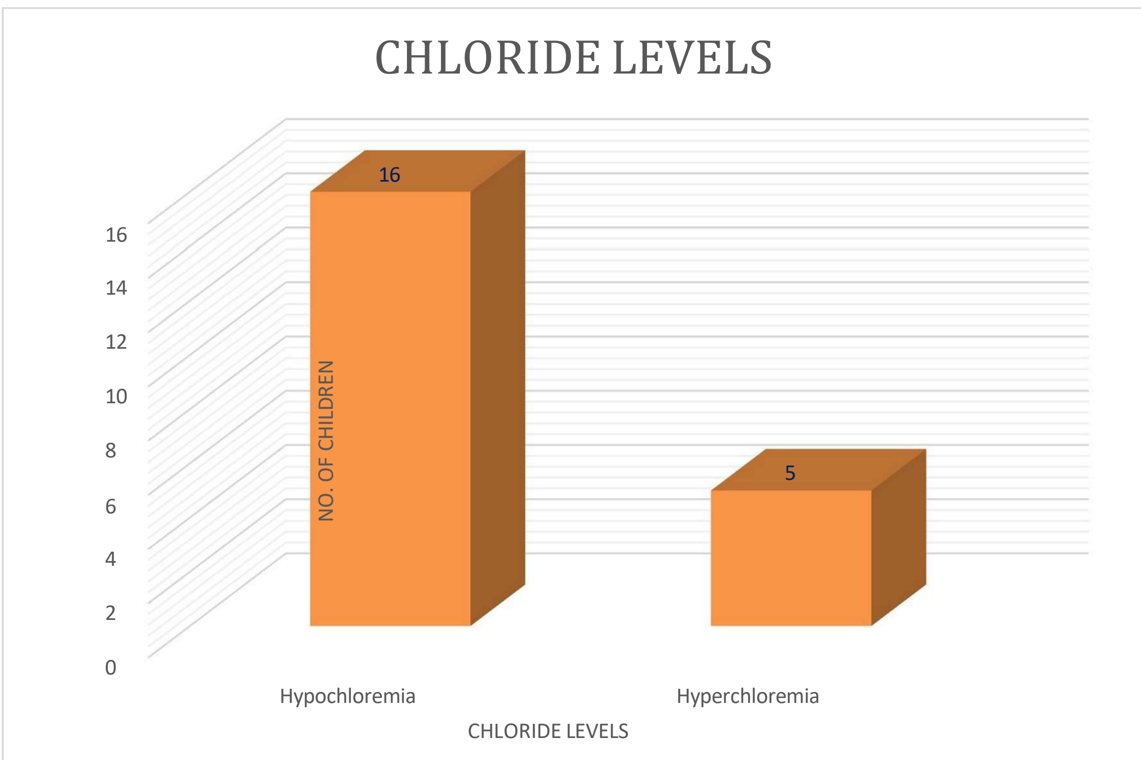

FIGURE.6.8: Distribution based on chloride levels

Out of 60 patients, 16 had hypochloraemia, whereas 5 had hyperchloremia

6.8 : DISTRIBUTION BASED ON STUNTING METHOD

TABLE 6.9 STUNTING METHOD

Percentages	No of children
<60%	2
70-80%	32
90-100%	26

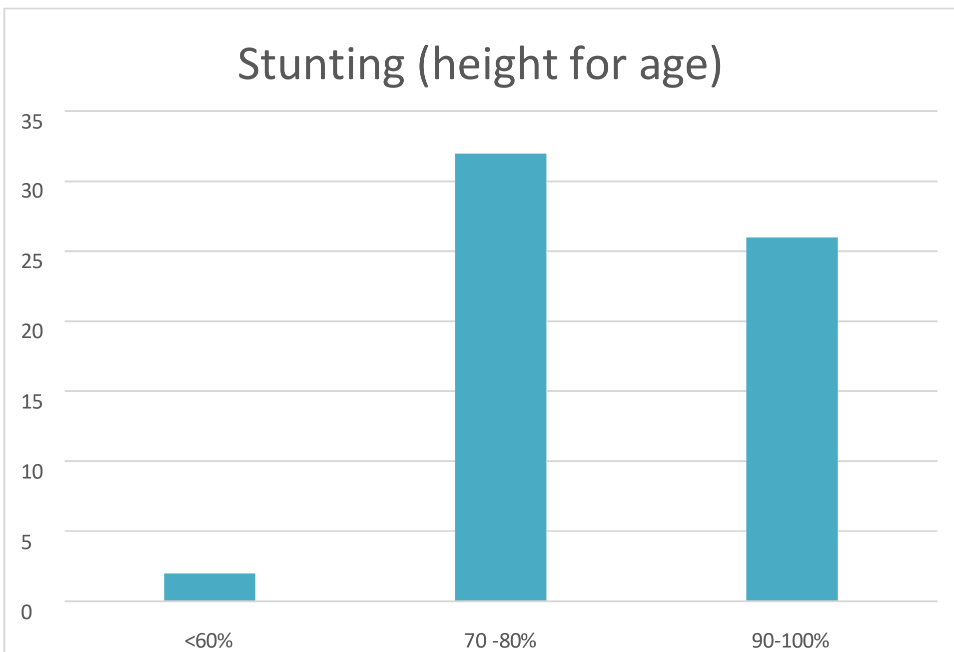

FIGURE.6.9: Distribution based on stunting method

Out of 60 patients, two had SAM. The remaining 26 were normal, and 32 were MAM

6.9 : DISTRIBUTION BASED ON WASTING METHOD

TABLE 6.1O WASTING METHOD

Percentages	No of children
<60%	9
70-80%	48
90-100%	3

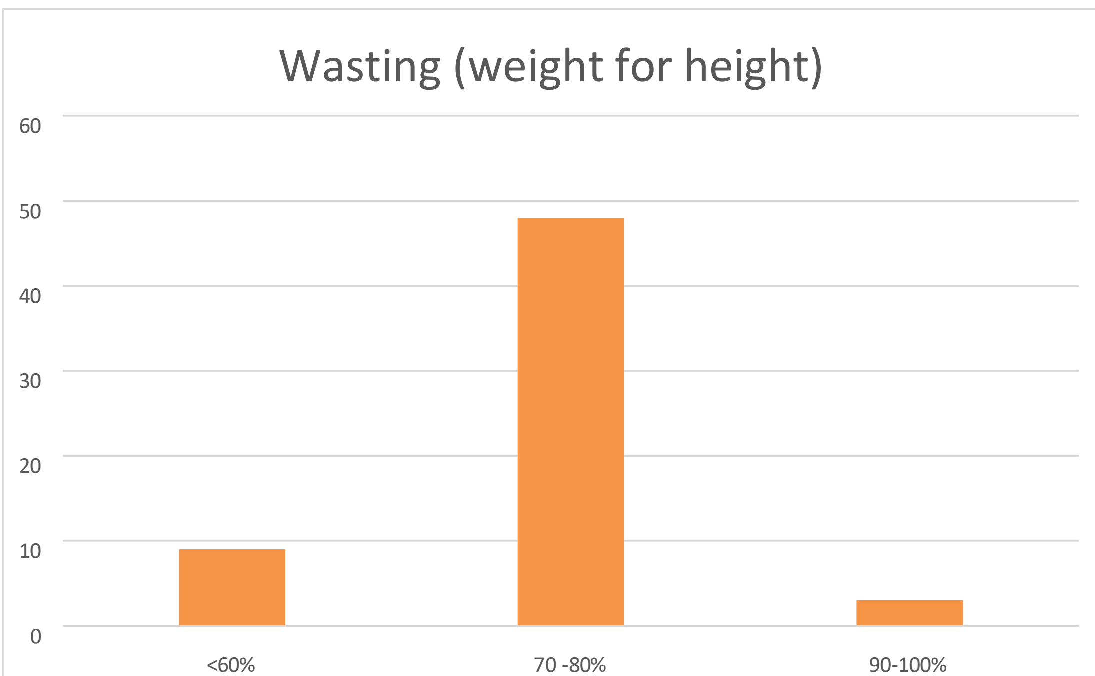

FIGURE.6.10: Distribution based on wasting method

9 out of 60 had SAM, 48 had MAM, and the remaining 3 had normal

6.10 : DISTRIBUTION BASED ON UNDER WEIGHT

TABLE 6.11 UNDER WEIGHT

percentages	No of children
<60%	37
70-80%	19
90-100%	4

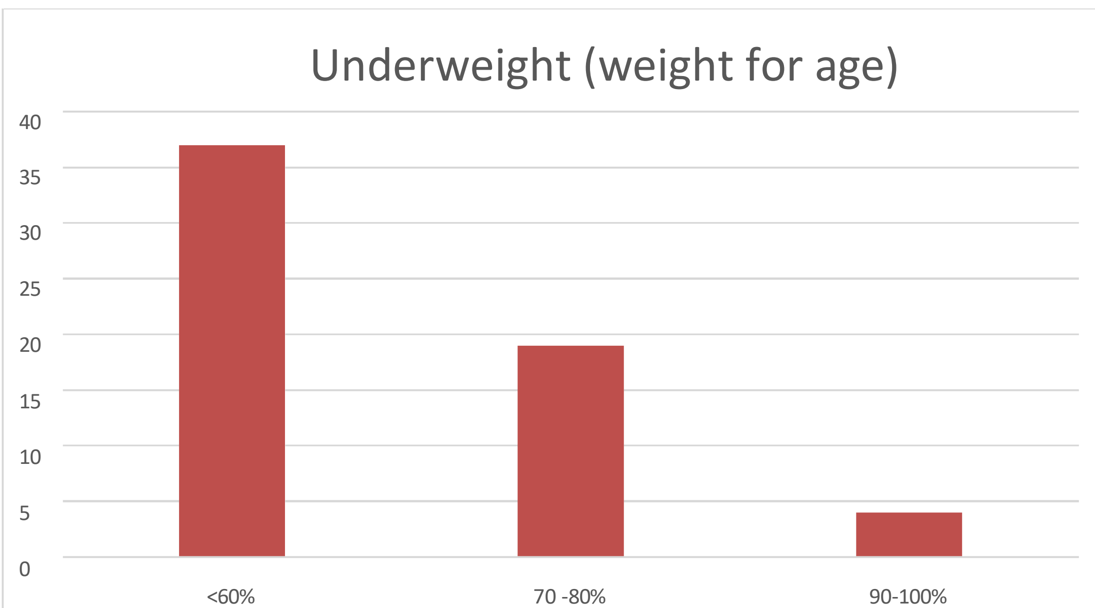

FIGURE.6.11: Distribution based on under weight

37 SAM, 19 MAM, and the remaining 4 participants in this study were normal.

6.11 DISTRIBUTION BASED ON AGE.

TABLE 6.12 AGE WISE DISTRIBUTION

AGE [IN YEARS]	FREQUENCY
0-1 YEAR	21
1-2 YEARS	21
2-3 YEARS	7
3-4 YEARS	8
4-5 YEARS	3

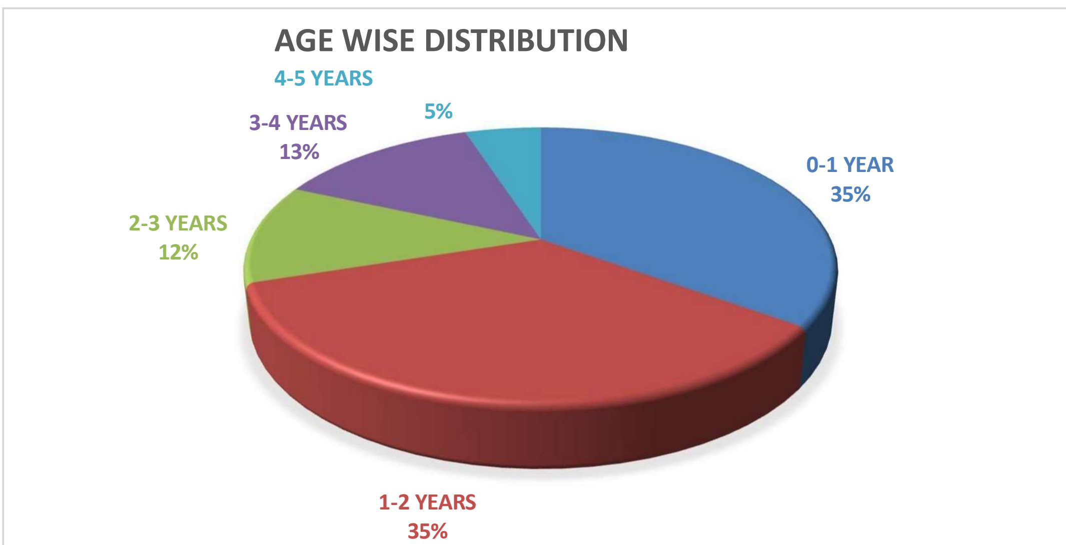

FIGURE.6.12: Age wise distribution

Out of 60 subjects, the majority of patients (35 %) were under the age group of 0-1years (n=21) and1-2 years (n=21), followed by the age group of 3-4 years constituting 13.33% (n=8). The least from the age group of 4-5 years constituting 5% (n=3).

DISCUSSION

- Current study was intended to explain the electrolytes profile with malnutrition, admitted in tertiary care hospital and their age were 6 to 60 months. This age group is very critical group because weakening is initiated and children have higher risk of developing malnutrition. There are 43.33% females, 56.66% males.
- Current results were comparable with the previous study conducted in India by Sadia khan et., al (1st March 2018 to 30th August 2018 at nutrition stabilization center of the children's hospital). Results were closely similar to our Results as they got 60% of males and 40% of females in the study. It may also indicate that despite Gender discrimination, male children were at more risk of malnutrition than female child, per phase more male children were admitted to hospital for treatment than female child.
- Current study indicates that the stunting method results that more children at the 80%Which indicates the pre moderate acute malnutrition and wasting method results indicates the more children at the 70% and 80% which knowns the moderate acute malnutrition and underweight calculation indication the more children seen in the 60% which knowns the severe acute malnutrition.
- In support to our study (Tran zika Gassmann et al conducted a study July 19 ,2022 from Latin America and Caribbean region exhibit some of the lowest undernutrition rate globally. From their study the results indicate that the wasting and stunting have reached low aggregate levels. Which shows the countries exist by residence and wealth states, with those in rural areas and in the poorest nutritional outcomes and complementary feeding practices.
- This study revealed that children suffering from severe acute malnutrition had aberrant serum electrolyte levels, with 10% having hyponatremia, 11% hypokalemia, and 16% hypochloremia. A study carried out in 2023 in the pediatric department of Nishtar Hospital in Multan provided support for our research. Dr. Arif zulqamain et al. It observed that grade III malnourished children exhibited the highest levels of electrolyte abnormalities.
- From our study the highest percentage is seen in 6months to 2years children with 35%, then 2-3years shows 12% then 3-4years shows 13%and 4-5years shows5% In support to our results the article Mohd Zakir Mohiuddin Owaisi et:al (A study of serum

electrolytes in malnourished children) on March 2020 the results include in 1-2years children shows 74%then 2-3of age shows 12%then 3-4age shows 8%and 4-5years age shows 6%.

- In support to our results (The article Out of 80 children studied, 44 (55%) children were aged below 12 months, 20 (25%) children were between 13 to 36 months,) and 16 (20%) children were between 3 to 5 years)STUDY OF ELECTROLYTE IMBALANCE IN CHILDREN SUFFERING FROM ACUTE GASTROENTERITIS OF UNDER 5 AGE GROUP Pedada Pratima 1,

SUMMARY

- The present study entitled "A prospective randomized study of different Assessment of Frequency of Serum Electrolytes (Na+, K+, Cl-) Disturbance in Malnourished Children with Diarrhea in a tertiary level teaching hospital" was conducted for a period of 6 months at Rajiv Gandhi Institute of Medical Sciences (RIMS) Hospital, Kadapa.
- The primary objective of our study was to assess the serum electrolyte abnormalities in malnourished children with diarrhea and also assess the grade of malnutrition.
- Malnutrition involves an imbalance between the availability of nutrients and energy and their requirements by the body for normal growth and body tissue functions. Severe malnutrition not only increases morbidity and mortality but it can also affect physical growth as well as psychological and intellectual development such as delayed motor skills acquisition and brain development. Malnutrition interacts with diarrhea in a vicious circle leading to high morbidity and mortality in children, and is a complicating factor for other illness in developing countries.
- Diarrhea is defined as the rapid transit of gastric contents through the bowel. According to WHO, diarrhea describes as three or more loose or watery stool per day. The absorption and secretion of water and electrolytes into the gut is a finely balanced, dynamic process.
- In our study, we included 60 patients they have Severe acute malnutrition (SAM) and Moderate acute malnutrition (MAM) and also electrolyte disturbances (Acute gastritis) after testing electrolyte profile and stunting, wasting and underweight
- In our study we concluded that, moderate acute malnutrition affected most of the children. Additionally, there were more boys than girls among the children. AND the majority of the kids suffered from grade 2 malnutrition.

IN THE VIEW OF ABOVE RESULT, WE CONCLUDE THAT

This study concludes that serum electrolytes disturbances in malnourished children are obvious during diarrheal illness particularly in those patients with Grade III Malnutrition irrespective of Age and duration of Diarrhea. Measurement of these Serum electrolytes is helpful for immediate therapy to avoid further complications.

RECOMMENDATION

- Additional research could support routine patient screenings for nutritional status.
- Further research could aid in gathering additional patient data for a more thorough analysis.

Sources

Sr. No	URL	Color	Percentage
1	https://pmc.ncbi.nlm.nih.gov		17.702%
2	https://www.scribd.com		11.119%
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